MODULE 13

Deep Learning

Build the intelligence behind image, language and recognition systems

LEARN A SKILL • BUILD PROOF • CREATE YOUR NEXT OPPORTUNITY

Understand neural networks from fundamentals through CNNs, sequence models and transformers, then turn concepts into projects.

WHO CAN JOIN?

AI/ML learners, students, freshers, Python learners, career switchers and technically curious non-IT learners.

WHY STUDENTS CHOOSE THIS PATH

- Beginner-friendly learning path — start with the fundamentals and progress step by step.
- Practical, project-focused training — learn by building instead of only watching lectures.
- Career-oriented skills — connect every topic to real applications and workplace use.
- Portfolio advantage — finish with projects you can discuss in interviews.
- A new direction is possible — your previous degree does not have to decide your entire career.

MODULE CONTENT

Linear algebra	Vectors/matrices
Probability	Statistics
Gradients	Neurons
Perceptron	Network architecture
Forward propagation	Backpropagation
Activation functions	Loss functions
Optimizers	Epochs
Batch size	Learning rate
Overfitting	Regularization
Dropout	Batch normalization
CNN	Convolution
Pooling	Image classification
Object detection	Segmentation
RNN	LSTM
GRU	Attention
Self-attention	Positional encoding
Encoder/decoder	TensorFlow
Keras	PyTorch

HANDS-ON PROJECTS

- Image Classifier

- Face Recognition
- Plant Disease Detection
- Object Detection
- Sentiment Analysis
- Handwritten Digit Recognition

CAREER DIRECTIONS

Deep Learning Engineer Beginner • Computer Vision Developer • AI/ML Engineer Beginner

YOUR JOURNEY

Learn → Practice → Build → Create a portfolio → Prepare for interviews → Apply your skills in real-world situations.

Note: Training improves skills and employability; employment depends on individual preparation, portfolio quality, interviews and available opportunities.